40. A cubic equation with real coefficients $p(x) = 0$ has one real root (α) and two complex roots ($\beta \pm i\gamma$) represented by A, B and C on the argand plane respectively. Then
(A) roots of $p'(x) = 0$ are complex if A lies inside one of the two equilateral triangles with base BC
(B) roots of $p'(x) = 0$ are real if A lies inside one of the two equilateral triangles with base BC
(C) roots of $p'(x) = 0$ are complex if A, B, C are vertices of an isosceles (not equilateral) triangle
(D) roots of $p'(x) = 0$ are real and equal if $\frac{\alpha-\beta}{\gamma} = \pm\sqrt{3}$

41. Let $A = \begin{bmatrix} 1 & 3 & 0 \\ 0 & 3 & 0 \\ 0 & -2 & 3 \end{bmatrix}$ be a given matrix. If $B = A^4 - 4A$ and $C = 15A^2 - 7A^3 - A$, then (Tr (M) and Det (M) represents trace and determinant of matrix M respectively)

(A) $Tr(B + C) = 28$
(B) $Tr(B + C) = 14$
(C) $Det(Adj(B + C)) = 576$
(D) $\sqrt{Det(Adj(Adj(B+C)))} = (576)^2$

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

42. If 6 digit numbers are formed using the digits 1 and 8 only and n_1, n_2, n_3 are the numbers of such 6 digits numbers which are divisible by 231, 84 and 63 respectively, then $n_3 - n_2 - n$, is

43. If $\int \frac{(e^x-1)\sin x - (e^x-1-x)\cos x}{1+(e^x-1-x-\cos x)(e^x-1-x+\cos x)} = f[g(x)] + c$ where c is constant of integration and $\lim_{x\to 0} g(x) = 0$. If $\lim_{x\to 0}\left(\frac{f(x)}{x} + \frac{g(x)}{x}\right) = \frac{m}{n}$ where m, n $\in$ N, then the minimum value of m + n is

44. Let ABCD be a square with vertices A(0, 0), B(4, 0), C(4, 4), D(0, 4). The square is folded in such a way that corner B always lies on AD. As B moves on AD, the crease thus formed always touches a fixed parabola whose length of latus rectum is

45. Five cards are drawn from a pack of 52 cards. If the probability of getting exactly 3 Diamonds and 2 Jack is P, then the value of $\frac{866320}{649}P$ is

46. Let $I = \int_0^1 \frac{(x^9 - x^5 + x)}{(3x^8 - 4x^4 + 6)^{\frac{3}{4}}} dx$, then the value of $(12\ I)^4$ is

47. The square of the product of reciprocal of roots of the equation $2\cot^{-1}(2x-1) + \sin^{-1}x = \pi$ is